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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/908,320	10/07/2024	Yu Fang	SONA-0006-U01-C01-C02	5680

87084 7590 03/25/2026

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EXAMINER	
PENA-SANTANA, TANIA M	

ART UNIT	PAPER NUMBER
2443	

NOTIFICATION DATE	DELIVERY MODE
03/25/2026	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

uspatents@gtclawgroup.com

Office Action Summary	Application No. 18/908,320	Applicant(s) Fang et al.	
	Examiner TANIA M PENA-SANTANA	Art Unit 2443	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 10/07/2024.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-16 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-16 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 10/07/2024 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date 01/07/2025.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

Notice of Pre-AIA or AIA Status

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Claims Status

Claims 1-16 are pending and have been rejected.

Information Disclosure Statement

The information disclosure statement (IDS) submitted on 01/07/2025 is in compliance with the provisions of 37 CFR 1.97. Accordingly, the information disclosure statement is being considered by the examiner.

Claim Rejections - 35 USC § 103

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

The factual inquiries for establishing a background for determining obviousness under 35 U.S.C. 103 are summarized as follows:

1. Determining the scope and contents of the prior art.

2. Ascertaining the differences between the prior art and the claims at issue.

3. Resolving the level of ordinary skill in the pertinent art.

4. Considering objective evidence present in the application indicating obviousness or nonobviousness.

Claims 1-16 are rejected under 35 U.S.C. 103 as being unpatentable over Roese et al. (U.S. Publication 2009/0187968), hereinafter 'Roese' in view of Kim et al. (U.S. Publication 2016/0255154), hereinafter 'Kim'.

As to claims 1 & 9, Roese discloses a method and an apparatus, comprising:

interpreting a policy comprising a network usage permission description (Roese, see [0012] & [0023], dynamic policy system can enable a network administrator to establish comprehensive control of network services usage);

configuring a first network interface circuit in response to the network usage permission description (Roese, see [0026], the system is able to enforce established and generated policies, on an initial and continual basis, based on usage permission rules established by a network administrator. It can restrict usage of the network system and its services based on the attached function's characteristics, the particular connection point through which network infrastructure connection is established, and network system events related or unrelated to the attached function. See [0044], the policy system is configured to maintain and update the information associated with the attached functions and the network infrastructure of the network system in a centralized database, including the saved policies history); and

Roese is silent to regulating communications, using the first network interface circuit, between end points of a first network zone of a vehicle and end points of a second network zone of the vehicle.

However, Kim discloses regulating communications, using the first network interface circuit, between end points of a first network zone of a vehicle and end points of a second network zone of the vehicle (Kim, [0034], a vehicular security network device includes a first conduit, which is a sole network path connecting a first security zone to an external network. The first security zone can be construed as being a logical and/or physical zone including a plurality of functional elements of a vehicle. See [0040], a gate keeper is disposed on the conduit to control an access to the security zone by an external source. See [0043], the gate keeper configured to maintain security of the subzone by the security element of a higher level than the gate keeper).

Therefore, it would have been obvious to a person of ordinary skill in the art before the effective filing date of the claimed invention to modify Roese in view of Kim in order to further modify the method provides dynamic network policy management from the teachings of Roese with the method of preventing a security attack/threat on a vehicle network from the teachings of Kim.

One of ordinary skill in the art would have been motivated because it would allow to analyze a risk to security of a vehicle and provide a security countermeasure against the analyzed risk. (Kim – 0006).

As to claims 2 & 10, Roese in view of Kim discloses everything disclosed in claims 1 & 10. Roese further discloses wherein the network usage permission description further comprises an external data access description (Roese, see [0045], network services usage is permitted an attached function via a connection point deemed not to be inherently secure (e.g., an edge Switch port associated with an

external internet connection), the policing system can prompt the attached function to initiate an improved connection, such as a VPN, or can notify the attached function that supplemental restrictions apply while in the insecure area); and the method further comprises: configuring a gatekeeper interface circuit in response to the external data access description (Roese, see [0026], enforce established and generated policies based on usage permission rules established by a network administrator. It can restrict usage of the network system and its services based on the attached function's characteristics, the particular connection point through which network infrastructure connection is established, and network system events related or unrelated to the attached function. All policy sets can be directed to all network entry devices. The policy sets can be apportioned among the network entry devices and attached functions forced to particular connection points based on established sets of policies. It is contemplated that multiple policies will be applied to the connection points); and regulating communications, using the gatekeeper interface circuit, with an external device in response to the external data access description (Roese, see [0045], network services usage is permitted an attached function via a connection point deemed not to be inherently secure (e.g., an edge switch port associated with an external internet connection), the policing system can prompt the attached function to initiate an improved connection, such as a VPN, or can notify the attached function that supplemental restrictions apply while in the insecure area).

As to claim 3 & 11, Roese in view of Kim discloses everything disclosed in claims 2 & 10. Roese further discloses wherein the external data access description comprises

permissions associated with the external device (Roese, see [0045], network services usage is permitted an attached function via a connection point deemed not to be inherently secure (e.g., an edge switch port associated with an external internet connection), the policing system can prompt the attached function to initiate an improved connection, such as a VPN, or can notify the attached function that supplemental restrictions apply while in the insecure area).

As to claims 4 & 12, Roese in view of Kim disclosed everything disclosed in claims 2 & 10. Kim discloses wherein: the external data access description comprises permissions associated with a flow of the vehicle (Kim, [0040-0041], a gate keeper is disposed on the conduit to control an access to the security zone by an external source. A security zone includes functional elements (vehicles) connected to the outside of the security zone through a conduit. A gate keeper is disposed on the conduit to control an access to the security zone by an external source.); and

Roese further discloses regulating communications comprises regulating communications based on a flow association of a communicating one of the end points of the first network zone or second network zone (Roese, see [0023], the dynamic policy function includes the sub-functions of monitoring the network for triggers, decision making of whether to modify an assigned set of policies and enforcement of the assigned set of policies. The dynamic policy function module of any particular device of the infrastructure can include any one or more of the three identified sub-functions. See [0045], network services usage is permitted an attached function via a connection point deemed not to be inherently secure (e.g., an edge switch port associated with an

external internet connection), the policing system can prompt the attached function to initiate an improved connection, such as a VPN, or can notify the attached function that supplemental restrictions apply while in the insecure area).

As to claims 5 & 13, Roese in view of Kim disclosed everything disclosed in claims 2 & 10. Roese further discloses wherein: the external data access description comprises permissions associated with an application (Roese, see [0042], the set of policies evaluation and possible policy change or modification can occur per session, per port, per flow, per user, per attached function, per application sought, etc.), wherein regulating communications comprises: regulating communications based on an application association of a communicating device, the communicating device comprising one of: the external device, an end point of the first network zone, or an end point of the second network zone (Roese, see [0023], the dynamic policy function includes the sub-functions of monitoring the network for triggers, decision making of whether to modify an assigned set of policies and enforcement of the assigned set of policies. The dynamic policy function module of any particular device of the infrastructure can include any one or more of the three identified sub-functions. See [0045], network services usage is permitted an attached function via a connection point deemed not to be inherently secure (e.g., an edge switch port associated with an external internet connection), the policing system can prompt the attached function to initiate an improved connection, such as a VPN, or can notify the attached function that supplemental restrictions apply while in the insecure area).

As to claims 6 & 14, Roese in view of Kim disclosed everything disclosed in claims 2 & 10. Kim further discloses wherein: the external data access description comprises permissions associated with a network zone of the vehicle, and regulating communications comprises: regulating communications based on at least one of a source zone or a destination zone of a regulated communication, and the external data access description (Kim, [0034], a vehicular security network device includes a first conduit, which is a sole network path connecting a first security zone to an external network. The first security zone can be construed as being a logical and/or physical zone including a plurality of functional elements of a vehicle. See [0040], a gate keeper is disposed on the conduit to control an access to the security zone by an external source. See [0043], the gate keeper configured to maintain security of the subzone by the security element of a higher level than the gate keeper).

As to claims 7 & 15, Roese in view of Kim disclosed everything disclosed in claims 6 & 14. Kim further discloses wherein the first network zone and the second network zone each comprise separate virtual local area networks of the vehicle (Kim, see [0100], the user uses application of the smartphone to communicate signal to the vehicle through a 3G/4G network or through the Internet, wherein the signal can access some of the functional elements through an internal network of the vehicle).

As to claims 8 & 16, Roese in view of Kim disclosed everything disclosed in claims 1 & 9. Roese discloses wherein the network usage permission description further

comprises: a network utilization description corresponding to at least one of the first network zone or the second network zone (Roese, see [0045], an attached function is permitted network services usage via a connection point deemed not to be inherently secure (e.g., an edge Switch port associated with an external internet connection), the policing system can prompt the attached function to initiate an improved connection, such as a VPN, or can notify the attached function that supplemental restrictions apply while in the insecure area.); and wherein regulating communications comprises regulating communications based on the network utilization description, and an associated communicating device associated with a regulated communication (Roese, see [0023], the dynamic policy function includes the sub-functions of monitoring the network for triggers, decision making of whether to modify an assigned set of policies and enforcement of the assigned set of policies. The dynamic policy function module of any particular device of the infrastructure can include any one or more of the three identified sub-functions. The policy sets can be apportioned among the network entry devices and attached functions forced to particular connection points based on established sets of policies. See [0045], network services usage is permitted an attached function via a connection point deemed not to be inherently secure (e.g., an edge switch port associated with an external internet connection), the policing system can prompt the attached function to initiate an improved connection, such as a VPN, or can notify the attached function that supplemental restrictions apply while in the insecure area)).

Kim further discloses and a communicating device description comprising at least one of an end point of one of the first network zone or the second network zone, a flow,

a vehicle function, or an application (Kim, see [0040], a security zone include functional elements having a same required security level, and is connected to the outside of the security zone through a conduit. A gate keeper is disposed on the conduit to control an access to the security zone by an external source. See [0100], smartphone application of user (vehicle owner) can provide communication in order to access a functional element for controlling the vehicle);

Conclusion

The prior art made of record and not relied upon is considered pertinent to applicant's disclosure. This includes:

U.S. Publication 2015/0365352, which describes managing capacity in a virtualized network.

U.S. Publication 2018/0309626, which describes virtualized network function decomposition.

Any inquiry concerning this communication or earlier communications from the examiner should be directed to TANIA M PENA-SANTANA whose telephone number is (571)270-0627. The examiner can normally be reached Monday - Friday 8am to 4pm EST.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Nicholas R Taylor can be reached at 5712723889. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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/TANIA M PENA-SANTANA/
Examiner, Art Unit 2443

/CHRISTOPHER B ROBINSON/
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